

ECON 1550: International Finance

Exchange Rates and the Foreign Exchange Market: An Asset Approach

A model of exchange rate determination

Exogenous variables

Variable	Description
R	Domestic interest rate
R^*	Foreign interest rate
E^e	Expected exchange rate

Endogenous variables

Variable	Description	Equation	Type of equation
E	Exchange rate	$R = R^* + \frac{E^e - E}{E}$	Equilibrium condition

A model of exchange rate determination

- Two investment opportunities
 - Domestic bond with return $R_{\$}$ in Dollars
 - Foreign bond with return R^* in Euros
- To be indifferent between the two investments, the **uncovered interest parity condition** must hold:

$$(\text{UIP}): R_{\$} = R^* + \frac{E_{\$/\text{EUR}}^e}{E_{\$/\text{EUR}}} - 1$$

Exchange Rates

Bonds

Realized vs Expected Returns

Expected Returns of Foreign Bond

Approximations

Uncovered Interest Parity

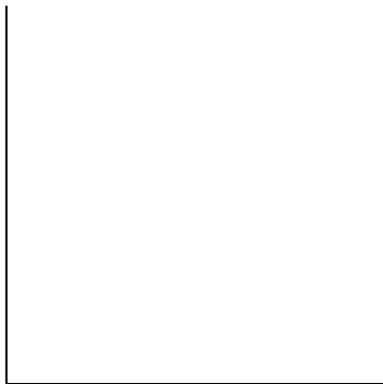
- For both strategies to have the same expected return

$$1 + R_{\$} = \frac{E_{\$/\text{EUR}}^e}{E_{\$/\text{EUR}}} (1 + R^*)$$

- Approximating

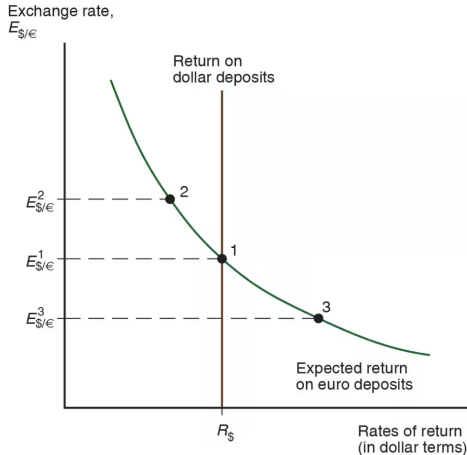
$$R_{\$} = R^* + \frac{E_{\$/\text{EUR}}^e}{E_{\$/\text{EUR}}} - 1$$

Equilibrium in Foreign Exchange Market



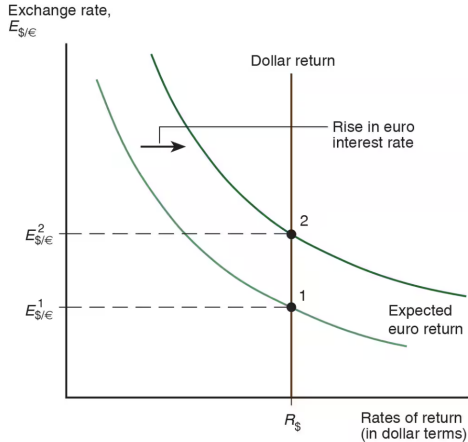
$$\text{UIP: } R_{\$} = R^{*} + \frac{E_{\$/\text{EUR}}^e}{E_{\$/\text{EUR}}} - 1$$

Equilibrium in Foreign Exchange Market



$$\text{UIP: } R_{\$} = R^{*} + \frac{E_{\$/\text{EUR}}^e}{E_{\$/\text{EUR}}} - 1$$

Shocks: Rise in Euro Interest Rate



The carry trade

- Borrowing at “low” rate R and lending at “high” rate R^* is a **carry trade**

$$\begin{array}{l} \text{expected return} \\ \text{on carry trade} \end{array} = R^* + \left(\frac{E^e}{E} - 1 \right) - R + \text{risk premium}$$

- Risk: Future exchange rate is not known when we start the carry trade, E^e can be different from the realized future exchange rate

Basic Carry Trade Returns

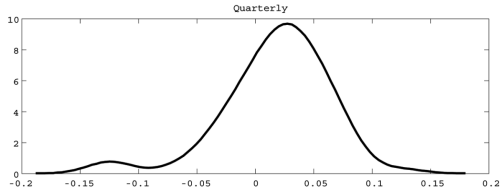
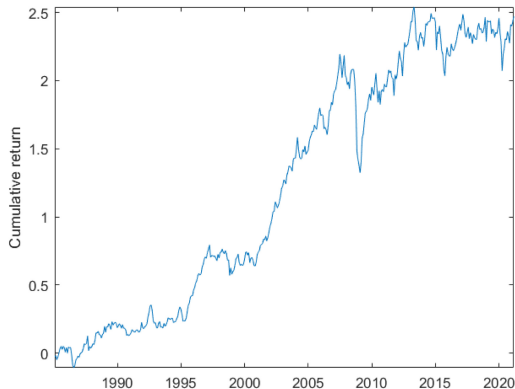


Fig. 1. U.S. dollar/Japanese yen exchange rate from 1996 to 2000

Source: Markus K. Brunnermeier, Stefan Nagel, Lasse H. Pedersen, "Carry Trades and Currency Crashes"

Basic Carry Trade Returns

“Conventional” carry trade



Source: Ho Choi, Sangwon Suh, “Conditionally-hedged currency carry trades”

Define the forward discount by:

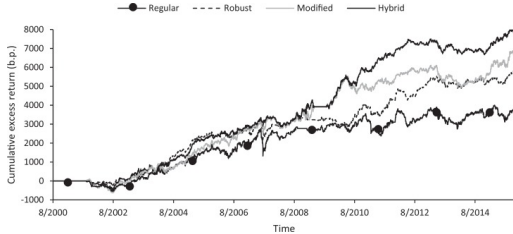
$$FD = E \frac{1 + R^*}{1 + R}$$

The carry trade returns on the plot are from a portfolio formed every month by lending in the three currencies with the highest forward discount and borrowing in the three currencies with the lowest forward discount, from the following set: Australia (AUD), Canada (CAD), Swiss franc (CHF), Euro (EUR) United Kingdom (GBP), Japan (JPY), Norway (NOK), New Zealand (NZD), Sweden (SEK)

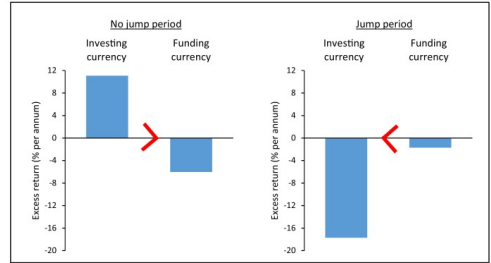
Explanations For Carry Risk Premium

- Crash risk / peso problem

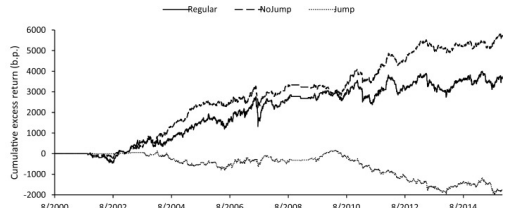
Effect of Jumps



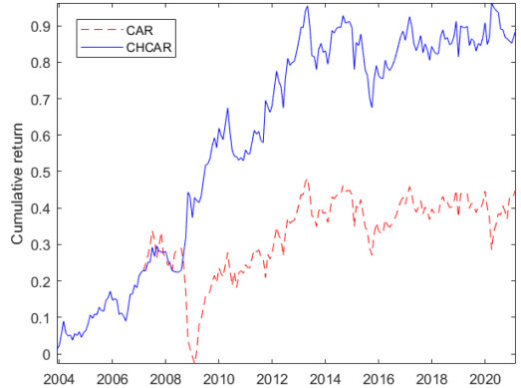
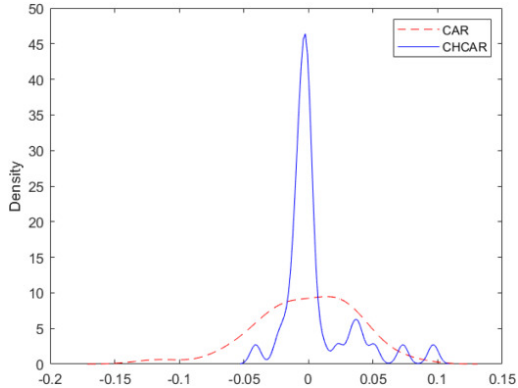
Panel A: Change in excess return with and without jumps



Panel B: Cumulative excess returns of currency carry trades



Avoiding Jumps



Source: Ho Choi, Sangwon Suh, "Conditionally-hedged currency carry trades"

Explanations For Carry Risk Premium

- Crash risk / peso problem
 - Infrequent large losses
 - Even if never before observed (peso problem)
- Correlation with consumption
 - Carry trade performs poorly when you need the money the most (recessions)

Higher Returns for Carry that Moves with Consumption

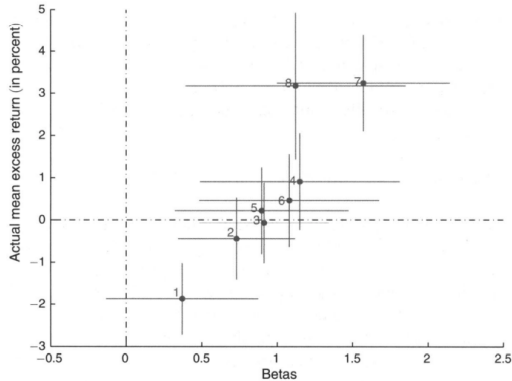


FIGURE 3. DURABLE CONSUMPTION GROWTH BETAS AND AVERAGE CURRENCY EXCESS RETURNS

Explanations For Carry Risk Premium

- Crash risk / peso problem
 - Infrequent large losses
 - Even if never before observed (peso problem)
- Correlation with consumption
 - Carry trade performs poorly when you need the money the most (recessions)
- Liquidity

ECON 1550: International Finance

Money, Interest Rates, and Exchange Rates

A model of the money market

Exogenous variables		Endogenous variables			
Variable	Description	Variable	Description	Equation	Type of equation
Y	Real income	R	Domestic interest rate	$M^d = P \times L(R, Y)$	behavioral equation
M^s	Money supply	M^d	Money demand	$M^d = M^s$	equilibrium condition
P	Price level				

Description of the Money Market

- There are only two assets: money and domestic bonds
- Money
 - Can be used for transactions
 - Pays no interest
- Bonds
 - Cannot be used for transactions

Money Demand

- Money demand is higher when:
 - Higher price level $P \Rightarrow$ need more money to buy goods
 - Nominal interest rate R is lower $\Rightarrow$ bonds less attractive
 - Income Y is higher $\Rightarrow$ more money to buy more goods
- Capture idea with a behavioral equation

$$M^d = P \times L\left(\underset{(-)}{R}, \underset{(+)}{Y}\right)$$

Real money demand

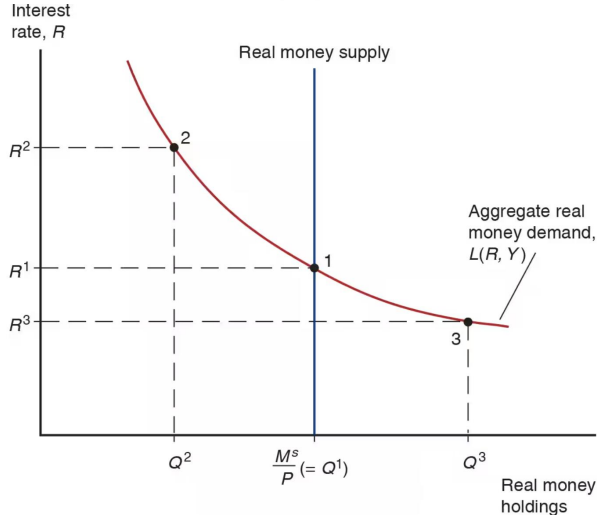
- Sometimes, convenient to write in terms of real money demand

$$\frac{M^d}{P} = L(R, Y)$$

and real money supply

$$\frac{M^s}{P}$$

Equilibrium in money market



Money is defined by its function

- Medium of exchange
- Store of value
- Unit of account

Many types of money

- Commodity money: a physical commodity (like gold) that is used as money
- Convertible paper money: a piece of paper that can be exchanged by a commodity
- Fiat money: issued by a central bank, not backed by any commodity

Fiat money

1. Central bank liabilities are special because they are the unit of account.
2. Currency is a promise to deliver future central bank liabilities.

Assets	Liabilities	Assets	Liabilities	Assets	Liabilities
\$0	\$0	\$1 currency	\$1 money	\$1 bonds	\$1 money
$t = 0$		$t = 1$		$t = 2$	

Table 1: Balance sheet of a central bank with fiat money. Initially, at time $t = 0$ (left panel), the central bank has zero assets and zero liabilities. At time $t = 1$ (center panel), the central bank creates one dollar of assets in the form of currency (a one dollar bill in paper money) and one dollar of liabilities in the form of money.

Monetary policy

- Deposits at the Fed are called reserves
- “The” interest rate is just interest on reserves